

The agile eye: a high-performance three-degree-of-freedom camera-orienting device

Clément M. Gosselin and Jean-François Hamel
Département de Génie Mécanique
Université Laval
Ste-Foy, Québec, Canada, G1K 7P4

Abstract

This paper presents some results obtained in the development of a three-degree-of-freedom camera-orienting device. The agile eye, as it is referred to, is capable of an orientation workspace larger than that of the human eye. The miniature camera mounted on the end-effector can be pointed within a cone of 140 degrees opening with plus or minus 30 degrees in torsion. The mechanical architecture of the orienting device is based on a spherical three-degree-of-freedom parallel manipulator which leads to high-performance dynamics. A kinematic optimization has been performed in order to determine the dimensional parameters of the prototype which would provide the best overall accuracy. A complete dynamical model of the manipulator has also been derived and programmed, and simulation results have guided the mechanical design. Finally, a prototype has been built and experimented.

1 Introduction

The design of a high-performance three-degree-of-freedom camera-orienting device is addressed in this paper. Applications for such a device are mainly associated with the tracking of objects moving at high speed using a camera. However, the mechanism developed here could also be used to control the orientation of mirrors, lasers or any device to be oriented precisely at high speed.

Camera-orienting devices have been proposed in the recent years [1], [2], some of which have been mounted on robotic heads. In [1] it has been pointed out that, in order to combine images of two cameras in a stereo pair, three degrees of freedom are required for the orientation of each of the cameras. Indeed, a small range of torsion of each of the cameras about its axis is necessary. This motion is accomplished, for example, by the human eyes [1]. Although the prototypes

of camera-orienting devices introduced so far have allowed to demonstrate the vision principles and have been used to implement tracking schemes [3], they have mostly been based on classical serial mechanical architectures.

In this paper, a parallel spherical three-degree-of-freedom architecture of manipulator is used as a motion base for the orienting device. Parallel mechanical architectures have been first introduced in simulation applications [4] and later on in robotics (see for instance [5], [6]). Parallel mechanisms are characterized by the fact that the end-effector is connected to the base via multiple kinematic chains and that all the actuators can be located on or close to the base. This leads to high stiffness and load-carrying capacity and to very good dynamic properties since the inertia of the moving parts is considerably reduced. Among other architectures, spherical parallel manipulators have received some attention ([7], [8]). More recently, families of isotropic spherical parallel manipulators have been obtained through a detailed kinematic analysis [9]. For the present application, an isotropic architecture that would have the best overall kinematic accuracy has been selected. The kinematic analysis and optimization of the manipulator will first be presented. Then, the dynamic model of the manipulator will be outlined and some simulation results will be presented. The mechanical design of the prototype will also be addressed and the constraints associated with the application will be discussed.

2 Kinematic analysis and optimization

The kinematic analysis of a spherical three-degree-of-freedom parallel manipulator and the solution of the inverse kinematic problem associated with this manipulator has been addressed in [7], [8]. The architecture of a general spherical three-degree-of-freedom

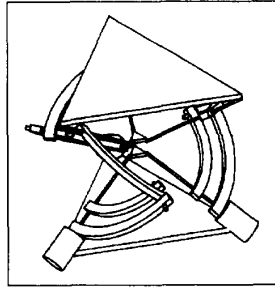


Figure 1: General architecture of a spherical three-degree-of-freedom parallel manipulator.

parallel manipulator is represented schematically in Fig. 1.

The structure of the manipulator is such that the axes of all the revolute joints intersect at one common point which is the *center of rotation* of the device. The three motors of the manipulator are fixed to the base and are represented as cylinders in the figure. The unit vectors directed along the axes of the actuators are referred to as vectors $\mathbf{u}_i$, $i = 1, 2, 3$ while the unit vectors directed along the axes of the revolute joints on the end-effector are referred to as vectors $\mathbf{v}_i$, $i = 1, 2, 3$. Moreover, the orientation of the end-effector with respect to the base is given by a rotation matrix $\mathbf{Q}$ and the actuator angles are noted θ_i , $i = 1, 2, 3$. The link angles of the manipulator are assumed to be the same on each of the legs connecting the end-effector to the base and are noted α_1 and α_2 . This is represented in Fig. 2 together with angles γ and β which define the geometry of the base and the end-effector links. It is clear, from the figure that angles γ and β are not independent. They are in fact related through the geometric constraint

$$\sin \beta = \frac{2\sqrt{3}}{3} \sin\left(\frac{\gamma}{2}\right) \quad (1)$$

and hence, only one of these angles suffices to define the geometry of the base or end-effector link. Finally, unit vectors directed along the axis of the intermediate revolute joint of each of the legs are noted $\mathbf{w}_i$, $i = 1, 2, 3$. Their expression is easily obtained as a function of the joint angles θ_i , $i = 1, 2, 3$ [7]. As shown in [7], [8], the solution of the inverse kinematic problem of this manipulator is straightforward and leads to a maximum of 8 solutions. The derivation of the solution to this problem is now briefly recalled. From Fig. 2, it is clear that the following equations hold,

$$\mathbf{w}_i \cdot \mathbf{v}_i = \cos \alpha_2, \quad i = 1, 2, 3 \quad (2)$$

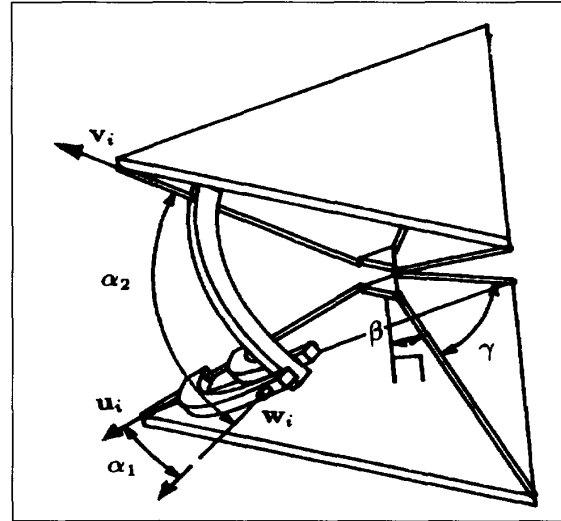


Figure 2: Geometric parameters of the spherical three-degree-of-freedom parallel manipulator.

Expressions of vectors $\mathbf{v}_i$ obtained from matrix $\mathbf{Q}$ and expressions of vectors $\mathbf{w}_i$ as functions of the actuated joint coordinates are then substituted in the above equations. This leads to three equations which can be written as

$$A_i T_i^2 + 2B_i T_i + C_i = 0, \quad i = 1, 2, 3 \quad (3)$$

with $T_i = \tan(\theta_i/2)$, $i = 1, 2, 3$ and where the coefficients A_i , B_i and C_i are functions of the Cartesian coordinates — i.e., the orientation of the end-effector — and of the geometric parameters of the robot. Hence, 2 solutions for each of the angles θ_i are obtained from eqs. (3) which leads to 8 solutions for the inverse kinematic problem.

Equations (2) can then be differentiated with respect to time in order to obtain the velocity equations. As shown in [8], this leads to an equation of the form

$$\mathbf{J}\boldsymbol{\omega} + \mathbf{K}\dot{\boldsymbol{\theta}} = \mathbf{0} \quad (4)$$

where $\boldsymbol{\omega}$ is the angular velocity of the end-effector and where vector $\boldsymbol{\theta}$ is the joint velocity vector defined as

$$\dot{\boldsymbol{\theta}} = [\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3]^T \quad (5)$$

Matrices $\mathbf{J}$ and $\mathbf{K}$ are referred to as the Jacobian matrices of the manipulator [8] and can be written, in invariant form, as

$$\mathbf{J} = \begin{bmatrix} (\mathbf{w}_1 \times \mathbf{v}_1)^T \\ (\mathbf{w}_2 \times \mathbf{v}_2)^T \\ (\mathbf{w}_3 \times \mathbf{v}_3)^T \end{bmatrix} \quad (6)$$

and

$$\mathbf{K} = \text{diag}(\mathbf{w}_1 \times \mathbf{u}_1 \cdot \mathbf{v}_1, \mathbf{w}_2 \times \mathbf{u}_2 \cdot \mathbf{v}_2, \mathbf{w}_3 \times \mathbf{u}_3 \cdot \mathbf{v}_3) \quad (7)$$

In a landmark paper, Salisbury and Craig [10] have defined the dexterity of a robotic manipulator as the kinematic accuracy associated with it. Mathematically, they defined the dexterity as the condition number of the Jacobian matrix of the robot. From the above velocity equation, the dexterity of the spherical three-degree-of-freedom parallel manipulator, noted κ , can then be written as

$$\kappa = \|\mathbf{J}^{-1}\mathbf{K}\| \|\mathbf{J}\| \quad (8)$$

where $\|\cdot\|$ denotes the Euclidean norm of its matrix argument, which is defined as

$$\|\mathbf{A}\| = \sqrt{\text{tr}(\mathbf{A}^T \mathbf{W} \mathbf{A})}; \quad \mathbf{W} = \frac{1}{n} \mathbf{1} \quad (9)$$

with n the dimension of the square matrix $\mathbf{A}$ and with $\mathbf{1}$ the $n \times n$ identity matrix. Moreover, one has

$$1 \leq \kappa < \infty \quad (10)$$

and hence, the reciprocal of the condition number, noted ζ , is used instead here. One has

$$\zeta = \frac{1}{\kappa}; \quad 0 \leq \zeta \leq 1 \quad (11)$$

Since indices κ and ζ are functions of the Jacobian matrices, they vary with the configuration of the robot and with the geometric parameters of the robot. A value of ζ equal or close to 1 corresponds to a configuration with a very good kinematic accuracy while a value of ζ equal to 0 is obtained when the manipulator is in a singular configuration. Manipulators which can attain at least one configuration where ζ is equal to 1 are referred to as *isotropic* manipulators and the corresponding configurations are termed *isotropic configurations*[10].

Since dexterity is a local criterion, i.e., a performance index which varies with the configuration of the manipulator, a *global conditioning index* has also been proposed in [11]. This index, noted η , is defined

$$\eta = \frac{\int_W \zeta dW}{\int_W dW} \quad (12)$$

where W denotes the workspace of the manipulator. The global conditioning index represents a measure of the overall kinematic performance of the manipulator and is a function of the architecture only.

Architecture	$\gamma(\text{deg.})$	$\alpha_1(\text{deg.})$	$\alpha_2(\text{deg.})$
A	90	90	90
B	120	120	90
C	105	90	105
D	83	90	90

Table 1: Some of the most promising architectures of isotropic spherical three-degree-of-freedom parallel manipulators.

In [9], families of isotropic spherical parallel manipulators have been obtained using the above expressions of the Jacobian matrices and the definition of the dexterity. In other words, isotropic loci in the design space of the robot — i.e., a space defined by the geometric parameters — have been obtained. This has allowed the identification of several isotropic architectures and the most promising of these architectures, which are listed in Table 1, have been investigated here.

First, a workspace analysis has shown that all the architectures listed in Table 1 can provide a workspace which satisfies the design constraints, i.e., pointing within a cone of 140° opening with $\pm 30^\circ$ in torsion. Then, computer programs have been written to investigate singularities and global conditioning. It has been shown that architecture C has a singularity when pointing in the reference direction with an angle of torsion of 20° . Hence, this architecture was eliminated. Physical dimensions have then been considered. Indeed, it is of utmost importance to avoid mechanical interference of the links throughout the workspace of the manipulator. Moreover, the dimensions of the manipulator should be kept as small as possible in order to minimize the inertia of the moving parts. Because of the link angles associated with them, architectures A and D can be built as two concentric spheres, one for the proximal links of each of the legs and one for the distal links. However, the longer link angles of the proximal links of architecture B do not allow such a simplification and require that all the links be located on concentric spheres. This would result in a larger prototype. Architecture B was therefore also eliminated. Finally, the global conditioning index and the minimum dexterity over the workspace have been considered. The results are given in Table 2 for architectures A and D for pointing only. Since architecture A has a better global conditioning and a larger minimum dexterity, it has been selected as the final kinematic architecture for the prototype. It is pointed out that,

Architecture	η	$\zeta_{\min}$
A	0,81	0,51
D	0,76	0,37

Table 2: Global conditioning and minimum dexterity over the target workspace (pointing only) for architectures A and D.

with no torsion, the worst configuration in the useful workspace leads to a conditioning of 0,51, which corresponds to a well-conditioned configuration. Undesirable configurations are therefore completely eliminated from the workspace.

3 Dynamic modeling

For the application envisioned here, it is desired to move the manipulator at high-speed with large accelerations. Hence, a dynamic analysis of the manipulator must be completed in order to determine the forces and moments which must be exerted by the actuators or resisted by the structural elements.

The dynamic analysis of a spherical three-degree-of-freedom parallel manipulator has been addressed in [7]. A similar approach has been adopted here and a complete dynamical model of the manipulator has been obtained using the Newton-Euler equations on each of the links. First, the kinematic model of the manipulator is used to determine the linear and angular accelerations of each of the links. Then the Newton-Euler equations are written for each of the 7 moving rigid bodies of the kinematic chain. This leads to 42 equations. However, as it was pointed out in [7], some simplifying assumptions must be made on the internal forces since the mechanism is overconstrained. The assumptions used in the latter reference have also been used here, i.e., it is assumed that the free-rotating joints connecting the proximal links to the distal links and the distal links to the end-effector cannot transmit axial forces. This finally leads to a linear system of 42 equations in 42 unknowns which is, however, rather sparse. The solution of this system leads to all the forces and moments at the joints, including the actuator torques. Detailed equations are cumbersome and are not given here because of space limitation.

A simulation program has been developed in order to provide a general tool for the dynamic analysis of spherical parallel manipulators of arbitrary archi-

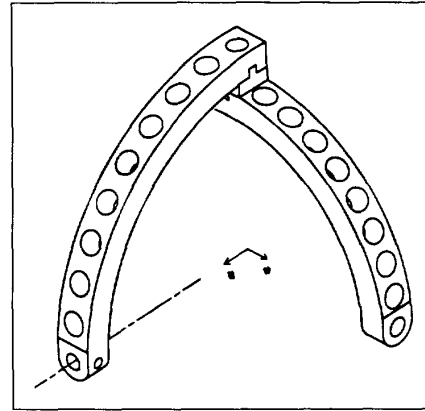


Figure 3: Proximal link of each of the legs.

ture. The program allows the simulation of any joint or Cartesian trajectory and the computation of the corresponding values of torques and forces at all joints and at the actuators. The simulation program has been used throughout the design of the manipulator to estimate the forces and torques to be expected for demanding trajectories.

4 Mechanical design

The kinematic architecture chosen for the manipulator requires a detailed analysis of the possible mechanical interferences. First, it is noticed that the proximal links — the links connected to the actuators — can interfere in most of the workspace because they can all move inside a pyramid formed by vectors u_i , $i = 1, 2, 3$. To overcome this problem, the proximal links have been designed as two circular arcs rigidly attached to each other, which does not affect the kinematic structure, but which minimizes the mechanical interferences. This is illustrated in Fig. 3 where a proximal link is represented. Some holes are drilled in the link in order to reduce the weight. Then, an exhaustive analysis of all types of possible interferences has been performed. It is recalled that the performance objectives in terms of workspace are a pointing cone of approximately 140° opening with $\pm 30^\circ$ in torsion. The different cases of possible mechanical interferences can be expressed in terms of geometric conditions. Angles whose value should be kept as large as possible are defined for each of these cases, namely: 1) the interference between the extremity of a distal link attached to the proximal link and the extremity attached to the end-effector of another distal link is

Torsion	$\mu_{\min}$	$\delta_{\min}$	$\lambda_{\min}$	$\epsilon_{\min}$
0	24,31	15,80	35,79	26,07
0 to 5	18,84	11,37	32,96	23,09
0 to 15	7,88	4,38	25,35	16,72
0 to 30	0,00	0,00	6,46	5,85

Table 3: Interference indices (deg.) for 140° pointing cone and given torsion angles.

measured by angle μ defined as

$$\mu = \cos^{-1}(\mathbf{w}_i \cdot \mathbf{v}_j), i \neq j \quad (13)$$

2) the interference between the extremity of a distal link attached to the proximal link and the body of another distal link is measured by angle δ defined as

$$\delta = \cos^{-1}[\sqrt{(\mathbf{w}_i \cdot \mathbf{v}_j)^2 + (\mathbf{w}_i \cdot \mathbf{w}_j)^2}], i \neq j \quad (14)$$

3) the interference between the vectors defined along the rigid joint connecting the two arcs composing a proximal link, for two different proximal links is measured by angle λ defined as

$$\lambda = \cos^{-1}[(\mathbf{u}_i \times \mathbf{w}_i) \cdot (\mathbf{u}_j \times \mathbf{w}_j)], i \neq j \quad (15)$$

4) finally, the interference between the axis of the camera and the vector along the joint of the two arcs composing a proximal link (which does not represent a mechanical interference but rather a case in which the links would appear in the cone of vision of the camera) can be measured by angle ϵ , defined as

$$\epsilon = \cos^{-1}[(\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3) \cdot (\mathbf{u}_i \times \mathbf{w}_i)] \quad (16)$$

where one has, in each of the above conditions, $i, j = 1, 2, 3$.

A computer program has been written in order to compute the above quantities and to determine the configuration in which they are minimal. Some results are given in Table 3. As shown in this table, maximum torsion cannot always be obtained near the boundary of the pointing workspace.

The most critical configurations obtained from the interference analysis have been studied which led to the design of the distal links. It was found that reducing the width of these links improved the workspace by reducing mechanical interferences. The resulting distal link is illustrated in Fig. 4.

Then, the passive revolute joints have been designed using miniature bearings in order to minimize the friction and the end-effector has been designed as a

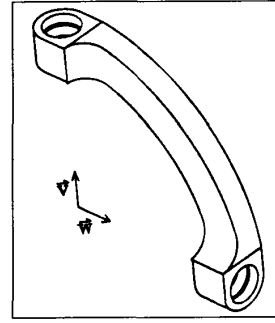


Figure 4: Distal link of each of the legs.

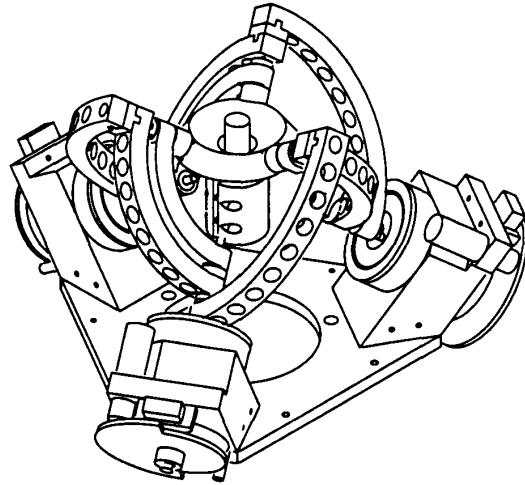


Figure 5: Complete design of the agile eye.

supporting device for a miniature commercially available camera. Finally, commercially available DC motors which can provide a maximum torque of 0,42 Nm and a maximum velocity 300 rad/s have been selected. Safety disks mounted on actuator shafts serve as mechanical limits to protect the mechanism and are used for the friction brakes as well as to protect the optical encoders mounted on each of the actuators. The complete design is represented in Fig. 5. A prototype has been built following this design. The prototype is shown in Fig. 6 and a video demonstrating the agile eye is also available. Preliminary experiments with the agile eye have been conducted and have shown that the performance of the manipulator are superior or equal to the design objectives. More advanced experiments are currently under way and advanced control schemes as well as tracking algorithms are being implemented.

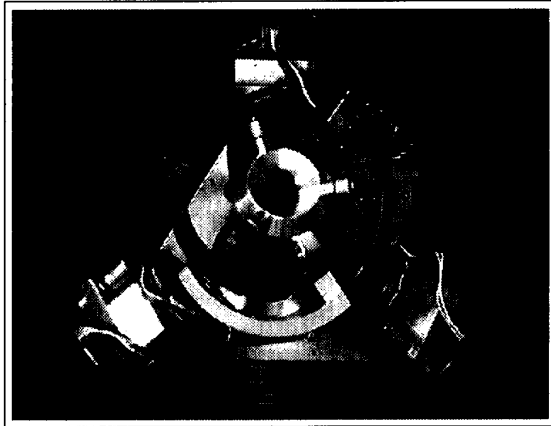


Figure 6: The agile eye: the prototype.

5 Conclusion

A high-performance camera-orienting device based on a spherical three-degree-of-freedom parallel manipulator has been proposed in this paper. First, a kinematic analysis has been performed and isotropic designs have been investigated. The optimization of the kinematic design has then been based on the global conditioning index and on the physical dimensions of the potential architectures. A dynamic analysis has also been performed and a complete simulation program has been written in order to determine the forces and torques for any given trajectory. Then a systematic mechanical interference analysis has been performed which allowed the determination of critical configurations and which led to the mechanical design of the links. Finally, a prototype has been built and experimented. Preliminary results have shown that the performances obtained are superior or equal to those expected. Future work includes the development of more advanced control and tracking algorithms.

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